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Started on Sunday, February 13, 2022, 9:48 PM

State Finished

Completed on Tuesday, February 22, 2022, 9:55 PM

Time taken 9 days

Points 32.50/60.00

Grade 54.17 out of 100.00



Question 1

Incorrect

0.00 points out of 10.00

As depicted in the figure below, a spherical particle settling through a quiescent fluid is subject to three forces: the downward force of gravity (F_G), and the upward forces of buoyancy (F_B) and drag (F_D). Both the gravity and buoyancy forces can be computed with Newton's second law with the latter equal to the weight of the displaced fluid. For laminar flow, the drag force can be computed with Stokes's law,

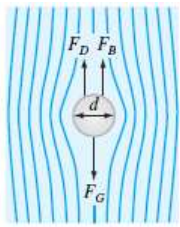
$$F_D = 3\pi\mu dv$$

where μ = the dynamic viscosity of the fluid (N s/m²), d = the particle diameter (m), and v = the particle's settling velocity (m/s). Note that the mass of the particle can be expressed as the product of the particle's volume and density ρ_s (kg/m³) and the mass of the displaced fluid can be computed as the product of the particle's volume and the fluid's density ρ (kg/m³). The volume of a sphere is $\pi d^3/6$ and laminar flow corresponds to the case where the dimensionless Reynolds number, Re, is less than 1, where $Re = \rho d v / \mu$.

In addition to the downward force of gravity (weight) and drag, an object falling through a fluid is also subject to a buoyancy force that is proportional to the displaced volume. For example, for a sphere with diameter d (m), the sphere's volume is $V = \pi d^3/6$ and its projected area is $A = \pi d^2/4$. The buoyancy force can then be computed as $F_b = -\rho V g$. We neglected buoyancy in our derivation of equation

$$\frac{dv}{dt} = g - \frac{c}{m} v$$

because it is relatively small for an object like a parachutist moving through air. However, for a more dense fluid like water, it becomes more prominent.



Question 1:

Identify a differential equation in the same fashion as equation

$$\frac{dv}{dt} = g - \frac{c}{m} v$$

but include the buoyancy force and represent the drag force.

Select one:

☒

$$m \frac{dv}{dt} = mg - \rho V g - \frac{1}{2} \rho v A C_d$$

✗

☐

$$m \frac{dv}{dt} = mg + \rho V g + \frac{1}{2} \rho v |v| A C_d$$

☐

$$m \frac{dv}{dt} = mg - \rho V g - \frac{1}{2} \rho v |v| A C_d$$

☐

$$m \frac{dv}{dt} = mg - \rho V g + \frac{1}{2} \rho v A C_d$$

↑



$$m \frac{dv}{dt} = mg - \rho V g + \frac{1}{4} \rho v A C_d$$

Question 2

Incorrect

0.00 points out of 10.00

Question 2:

Identify the differential equation from the previous problem of the same section for the special case of a sphere.

Select one:



$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{3\rho C_d}{4\rho_s d} v$$



$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) + \frac{3\rho C_d}{4\rho_s d} v$$



$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) + \frac{3\rho C_d}{4\rho_s d} v |v|$$



$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{3\rho C_d}{4\rho_s d} v |v|$$



$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{3\rho C_d}{8\rho_s d} v |v|$$



Question 3

Correct

10.00 points out of 10.00

Question 3:

Use the equation developed in Question 2 to determine the terminal velocity (i.e., for the steady-state case) with an initial velocity of zero.

Use the following parameter values for a sphere falling through water: sphere diameter = 1 cm, sphere density = 2700 kg/m^3 , water density = 1000 kg/m^3 , and $C_d = 0.47$.

Select one:

- ☐ 0.6783 m/s
- ☒ 0.68783 m/s
- ☐ 0.5783 m/s
- ☐ 0.78783 m/s
- ☐ 0.98783 m/s

**Question 4**

Correct

10.00 points out of 10.00

Question 4:

Use Euler's method with a step size of $\Delta t = 0.03125 \text{ s}$ to numerically calculate for the velocity from $t = 0$ to 0.25 s with an initial velocity of zero.

Select one:

- ☒ 0.683860
- ☐ 0.678859
- ☐ 0.667761
- ☐ 0.643887
- ☐ 0.623887



Question 5

Partially correct

2.50 points out of 10.00

The average daily temperature for an area can be approximated by the following function,

$$T = T_{\text{mean}} + (T_{\text{peak}} - T_{\text{mean}}) \cos(\omega(t - t_{\text{peak}}))$$

where T_{mean} = the average annual temperature, T_{peak} = the peak temperature, ω = the frequency of the annual variation ($= 2\pi/365$), and t_{peak} = day of the peak temperature ($\cong 205$ d).

Question 5

Match the unknowns A, B, C, and D to their corresponding values for the following the MATLAB code that computes the average temperature between two days of the year for a particular city.

```
function Ta = avgtemp(Tm,Tp,ts,te)
w = A
t = B
T = C
Ta = D
```

1. A
2. C
3. B
4. D

1	2*pi/365	✓
3	mean(T);	✗
2	ts:te;	✗
4	Tm + (Tp-Tm)*cos(w*(t-205));	✗

Question 6

Correct

10.00 points out of 10.00

Piecewise functions are sometimes useful when the relationship between a dependent and an independent variable cannot be adequately represented by a single equation. For example, the velocity of a rocket might be described by

$$v(t) = \begin{cases} 11t^2 - 5t & 0 \leq t \leq 10 \\ 1100 - 5t & 10 \leq t \leq 20 \\ 50t + 2(t - 20)^2 & 20 \leq t \leq 30 \\ 1520e^{-0.2(t-30)} & t > 30 \\ 0 & \text{otherwise} \end{cases}$$

Question 6

Match the unknown steps A, B, C, and D to their corresponding MATLAB codes to implement the piecewise function for the given problem.

The incomplete MATLAB function for the given problem is,

```
function v = vpiece(t)
if A
    v = 0;
elseif B
    v = 11*t^2 - 5*t;
elseif C
    v = 1100 - 5*t;
elseif D
    v = 50*t + 2*(t - 20)^2;
else
    v = 1520*exp(-0.2*(t-30));
end
```

1. A
2. B
3. C
4. D

t < 0	1	✓
t < 10	2	✓
t < 20	3	✓
t < 30	4	✓

◀ Quiz 0

Jump to...

Quiz 2 ▶

